mergencystretch

Chapter 4. Linear Structural Causal Models

Path coefficients, Wright's rules, single-door and instrumental criteria

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Motivation. The central question is when, in a linear SCM with Gaussian errors, a structural coefficient (a causal effect) can be computed from the observed covariance matrix, and when ordinary regression returns it.

Outline:

(1) linear SCMs and Wright's path-tracing rules;

(2) when regression can and cannot be used to find causal effects (single-door, back-door);

(3) algorithmic identification (systems of polynomial equations, instrumental variables, conditional IVs, instrumental sets), with appendices on complexity, regression in a reversed graph, and Gröbner bases.

4.1 Linear structural causal models

Motivation. Specialize SCMs to linear mechanisms and read covariances off the graph.

4.1.1 Definition and conventions

A linear SCM is a system of linear equations representing the ground truth,

$$Y = \sum_i \lambda_{x_i y} X_i + \mathcal{E}_y,$$

where the X_i are the direct causes of Y and $\lambda_{x_i y}$ is the change in Y per unit change of X_i. All covariances among the errors $\mathcal{E}$ are explicitly specified. Without loss of generality the data are normalized ($\mathbb{E}[X] = 0$, $\mathbb{E}[X^2] = 1$), and $\mathcal{E} \sim \mathcal{N}$, so the observed distribution is fully specified by the covariance matrix $\Sigma = (\sigma_{ij})$.

The only substantive change from the non-parametric case is that f_i becomes linear:

$$V_i \leftarrow f_i(\mathbf{pa}_i, \mathbf{u}_i) \quad \rightsquigarrow \quad V_i \leftarrow \sum_{V_j \in Pa_i} \lambda_{ji} V_j + \mathcal{E}_i.$$

(1) λ_{ji} is the *structural coefficient*;

(2) by convention U_i is renamed $\mathcal{E}_i$;

(3) knowing all λ_{ji} yields all causal effects.

Example 4.1

Non-parametric $X \leftarrow f_X(U_X)$, $W \leftarrow f_W(U_W)$, $Y \leftarrow f_Y(X, W, U_Y)$ becomes $X \leftarrow \mathcal{E}_x$, $W \leftarrow \mathcal{E}_w$, $Y \leftarrow \lambda_{xy} X + \lambda_{wy} W + \mathcal{E}_y$. Writing the coefficients on the edges of the graph fully specifies the model.

Latent confounding. The covariance $\epsilon_{ij} = \mathbb{E}[\mathcal{E}_i \mathcal{E}_j]$ is attached to a bidirected edge $V_i \leftrightarrow V_j$. It is unobserved, but drawn on the graph like a structural coefficient. Unlike the non-parametric case, a bidirected edge here represents *error covariance*, not necessarily a shared latent variable.

Interventions. For $X \xrightarrow{\lambda} Y$,

$$\mathbb{E}[Y \mid \mathrm{do}(X = x)] = \mathbb{E}[\lambda x + \mathcal{E}_y] = \lambda x + \mathbb{E}[\mathcal{E}_y] = \lambda x.$$

4.1.2 The identification problem

We are given (i) a hypothesized graph $\mathcal{G}$ (what causes what, and which pairs share unknown common causes), (ii) observational data $\{(x^{(i)}, y^{(i)})\}_{i=1}^N$ on all observed variables, but (iii) not the structural coefficients λ (the causal effects we want) nor the error covariances ϵ.

4.1.3 Connecting observed and unobserved quantities

For $X \xrightarrow{\lambda} Y$ without confounding,

$$\sigma_{xy} = \mathbb{E}[XY] = \mathbb{E}[X(\lambda X + \mathcal{E}_y)] = \lambda \underbrace{\mathbb{E}[XX]}_{1} + \underbrace{\mathbb{E}[X\mathcal{E}_y]}_{0} = \lambda.$$

With confounding $X \leftrightarrow Y$ (ϵ_{xy}), $X = \mathcal{E}_x$ and

$$\sigma_{xy} = \lambda\mathbb{E}[XX] + \mathbb{E}[X\mathcal{E}_y] = \lambda + \mathbb{E}[\mathcal{E}_x\mathcal{E}_y] = \lambda + \epsilon_{xy}.$$

A curious property. For the chain $X \xrightarrow{\lambda_{xz}} Z \xrightarrow{\lambda_{zy}} Y$:

$$\begin{aligned}\sigma_{xy} &= \mathbb{E}[X(\lambda_{zy}Z + \mathcal{E}_y)] = \lambda_{zy}\mathbb{E}[XZ] + \mathbb{E}[X\mathcal{E}_y] = \lambda_{zy}\mathbb{E}[X(\lambda_{xz}X + \mathcal{E}_z)] \\ &= \lambda_{zy}\lambda_{xz}\mathbb{E}[XX] + \lambda_{zy}\mathbb{E}[X\mathcal{E}_z] = \lambda_{zy}\lambda_{xz} + \lambda_{zy}\mathbb{E}[\mathcal{E}_x\mathcal{E}_z].\end{aligned}$$

Without confounding the last term vanishes: $\sigma_{xy} = \lambda_{zy}\lambda_{xz}$. If $X \leftrightarrow Z$ with covariance ϵ_{xz}, then $\sigma_{xy} = \lambda_{zy}\lambda_{xz} + \lambda_{zy}\epsilon_{xz}$.

4.1.4 Treks and Wright's rule

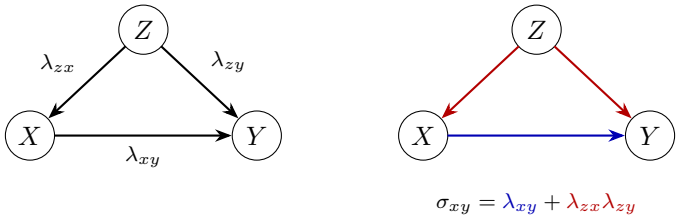


Figure. *Wright's rule for standardized variables: the covariance of X and Y is the sum, over the treks between them, of the products of path coefficients: the direct path (blue) and the trek through the common cause Z (red).*

For $Z \to X \to Y$ with $Z \to Y$: $\sigma_{xy} = \lambda_{xy} + \lambda_{zx}\lambda_{zy}$. The terms correspond to the paths $X \to Y$ and $X \leftarrow Z \to Y$.

Theorem 4.1 (Wright's rules, 1921)

For an acyclic linear SCM with standardized variables, σ_{xy} equals the sum, over all *open* paths between X and Y (non-self-intersecting paths without colliding arrowheads, i.e. treks; a bidirected edge counts as one step contributing ϵ), of the product of the path coefficients along the path. In particular $\sigma_{xy} = 0$ when X and Y are d-separated by $\emptyset$.

Proof [Supplement]. Write the model as $\mathbf{V} = \Lambda^{\mathsf{T}}\mathbf{V} + \boldsymbol{\epsilon}$ with Λ the matrix of path coefficients (strictly triangular in a topological order) and $\Omega = \mathrm{Cov}(\boldsymbol{\epsilon})$ (off-diagonal entries are the bidirected coefficients). Then $\mathbf{V} = (I-\Lambda^{\mathsf{T}})^{-1}\boldsymbol{\epsilon}$ and $\Sigma = (I-\Lambda)^{-\mathsf{T}}\Omega(I-\Lambda)^{-1}$. Since Λ is nilpotent, $(I-\Lambda)^{-1} = \sum_{k\geq 0}\Lambda^k$, and $(\Lambda^k)_{ij}$ is the sum over directed paths of length k from i to j of the products of coefficients. Hence

$$\sigma_{xy} = \sum_{a,b}\Big(\sum_{\text{directed } a\to x}\prod\lambda\Big)\,\omega_{ab}\,\Big(\sum_{\text{directed } b\to y}\prod\lambda\Big),$$

a sum over pairs of directed paths ending at x and y joined at their sources by either a common source $a = b$ (weight ω_{aa}, the error variance) or a bidirected edge $a \leftrightarrow b$ (weight ω_{ab}). These are exactly the treks between X and Y. With standardized variables, the error variance at the source of a trek through a common cause combines with the source's other contributions to give the variance

1 when the trek is read through the source node itself, which yields Wright's form with a factor 1 at the top of each trek. If $\emptyset$ d-separates X and Y, there are no treks and $\sigma_{xy} = 0$. ■

Consequently, if $X \xrightarrow{\lambda} Y$ is in the model, $\sigma_{xy} = \lambda +$ (other paths), and $\sigma_{xy} = \lambda$ if X and Y are d-separated in $\mathcal{G}_\lambda$ (the graph with that edge removed).

Example 4.2

$W \to X$, $W \to Y$, $Z \to W$, $Z \to X$, $X \to Y$: $\sigma_{xy} = \lambda_{xy} + \lambda_{wx}\lambda_{wy} + \lambda_{zx}\lambda_{wz}\lambda_{wy}$ (paths $X \to Y$, $X \leftarrow W \to Y$, $X \leftarrow Z \to W \to Y$).

Example 4.3

$W \to X$, $W \to V \to Y$, $X \to Z \to Y$, $X \leftrightarrow Z$: $\sigma_{xy} = (\lambda_{xz} + \epsilon_{xz})\lambda_{zy} + \lambda_{wx}\lambda_{wv}\lambda_{vy}$.

4.2 Linear regression and causal effects

Motivation. Clarify what regression computes and when it coincides with a structural coefficient.

4.2.1 The medical researcher

We want to know whether a new drug helps; our recommendation will be followed by doctors nationwide.

Step 1. gather data on how much drug patients took (X) and the amount of a biomarker (antibodies) in their blood (Y).

Step 2. regress $Y = \beta X + e$, giving $\beta = 0.375$; the drug seems beneficial and is authorized.

Step 3. given to everyone, the drug shows a clear *negative* relation with slope -1: it hurts. Not enough data is not the explanation. Why did regression fail (red line) and can the true causal effect (green line) be recovered?

4.2.2 The implicit assumption: no confounding

Attributing causal meaning to β presumes $X \leftarrow \mathcal{E}_x$, $Y \leftarrow \lambda_{xy}X + \mathcal{E}_y$ with $\mathcal{E}_x \perp\!\!\!\perp \mathcal{E}_y$; then $\sigma_{xy} = \lambda_{xy}\mathbb{E}[XX] + \mathbb{E}[X\mathcal{E}_y] = \lambda_{xy}$.

The ground-truth model. If one cannot ascertain the absence of confounding between X and Y, the corresponding model is $X \leftarrow \mathcal{E}_x$, $Y \leftarrow \lambda_{xy}X + \mathcal{E}_y$ with $\mathcal{E}_x, \mathcal{E}_y$ *correlated* ($X \leftrightarrow Y$). Then regression of Y on X is biased:

$$\sigma_{xy} = \lambda_{xy}\mathbb{E}[XX] + \mathbb{E}[\mathcal{E}_x\mathcal{E}_y] = \lambda_{xy} + \epsilon_{xy},$$

and it is provably impossible to disentangle the effect of the drug from the confounding: λ_{xy} is not identifiable. This explains the medical researcher's failure (0.375 versus -1).

What does regression compute? For $Y = \beta X + e$, minimize $\mathbb{E}[(Y - \beta X)^2] = 1 - 2\beta\sigma_{xy} + \beta^2$. The derivative $-2\sigma_{xy} + 2\beta$ vanishes at $\beta = \sigma_{xy}$: *the regression coefficient is just the covariance* (for standardized variables).

4.2.3 Regression equation versus structural equation

- **Regression equation** $Y = \beta X + e$ with $e \perp\!\!\!\perp X$ by construction. Solved, $\beta = \sigma_{xy}$, denoted r_{yx}. It makes *no* causal claim.
- **Structural equation** $Y \leftarrow \lambda X + \mathcal{E}_y$ (for $X \to Y$). It claims $\mathbb{E}[Y \mid \mathrm{do}(X = x)] = \lambda x$, which can be tested and falsified.

Confusing the two has been called the "confusion of the century". Causal claims via regression implicitly assume a model of the form $X_1 \xrightarrow{b_1} Y \xleftarrow{b_2} X_2$ (with the regressors unconfounded with Y), in which OLS of $Y = \beta_1 X_1 + \beta_2 X_2 + e$ gives $\beta_1 = b_1$, $\beta_2 = b_2$.

Warning: make sure you have the right graph. If instead Y causes both, $X_1 \xleftarrow{b_1} Y \xrightarrow{b_2} X_2$, simple regressions give $r_{yx_1} = b_1$, $r_{yx_2} = b_2$, but multiple regression gives

$$\beta_1 = \frac{b_1(1 - b_2^2)}{1 - b_1^2 b_2^2}, \qquad \beta_2 = \frac{b_2(1 - b_1^2)}{1 - b_1^2 b_2^2}$$

(derived in Appendix 4.6).

4.2.4 Testing conditional independence with regression

For Gaussian variables, $(X \perp\!\!\!\perp Y \mid Z)$ iff $r_{yx \cdot z} = 0$, the partial regression coefficient of X in $Y = \alpha X + \beta Z$. If $\alpha \neq 0$, Z does not render X and Y independent; if $\alpha = 0$, $(X \perp\!\!\!\perp Y \mid Z)$.

4.2.5 Single-door criterion: direct effects

Model: $Z \xrightarrow{\lambda_{zx}} X \xrightarrow{\lambda_{xy}} Y$, $Z \leftrightarrow Y$ (ϵ_{zy}). Regress $Y = \alpha X + \beta Z + e$:

$$\mathbb{E}[(Y - \alpha X - \beta Z)^2] = 1 - 2\alpha\sigma_{xy} - 2\beta\sigma_{yz} + 2\alpha\beta\sigma_{xz} + \alpha^2 + \beta^2,$$

$$\frac{\partial}{\partial \alpha} = 2\alpha - 2\sigma_{xy} + 2\beta\sigma_{xz}, \qquad \frac{\partial}{\partial \beta} = 2\beta - 2\sigma_{yz} + 2\alpha\sigma_{xz}.$$

Setting both to zero and substituting the Wright expressions ($\sigma_{xz} = \lambda_{zx}$, $\sigma_{xy} = \lambda_{xy} + \lambda_{zx}\epsilon_{zy}$, $\sigma_{yz} = \lambda_{xy}\lambda_{zx} + \epsilon_{zy}$):

$$\begin{aligned} \lambda_{xy} + \lambda_{zx}\epsilon_{zy} &= \alpha + \beta\lambda_{zx}, \\ \lambda_{xy}\lambda_{zx} + \epsilon_{zy} &= \alpha\lambda_{zx} + \beta, \end{aligned} \qquad \Longrightarrow \qquad \alpha = \lambda_{xy}, \quad \beta = \epsilon_{zy}.$$

(The linear system has a unique solution when $\lambda_{zx}^2 \neq 1$, and the stated values satisfy it.) So the coefficient of X is the causal effect, even though Z is confounded with Y.

Theorem 4.2 (Single-door criterion; Pearl 2000)

Let λ be the coefficient of $X \to Y$ and $\mathcal{G}_\lambda$ the diagram with $X \to Y$ removed. λ is identifiable if there is a set $\mathbf{Z}$ such that (i) $\mathbf{Z}$ contains no descendant of Y, and (ii) $\mathbf{Z}$ d-separates X from Y in $\mathcal{G}_\lambda$. Moreover $\lambda = r_{yx \cdot \mathbf{Z}}$.

Proof [Supplement]. Let β be the coefficient of X in the population regression of Y on X and $\mathbf{Z}$; it is the partial regression coefficient $r_{yx \cdot \mathbf{Z}}$. Write $Y = \lambda X + R$, where R collects all other parent contributions and the error of Y. Then $r_{yx \cdot \mathbf{Z}} = \lambda + r_{rx \cdot \mathbf{Z}}$, and $r_{rx \cdot \mathbf{Z}} = 0$ iff R and X are uncorrelated given $\mathbf{Z}$ (Gaussian case: conditionally independent). Every trek between X and R corresponds to a path between X and Y in $\mathcal{G}_\lambda$ (the edge $X \to Y$ removed); condition (ii) blocks all of them given $\mathbf{Z}$, and condition (i) ensures that conditioning on $\mathbf{Z}$ does not open a path through a descendant of Y (which would involve R itself). Hence $r_{rx \cdot \mathbf{Z}} = 0$ and $\lambda = r_{yx \cdot \mathbf{Z}}$. ■

Example 4.4

$A \to B$, $A \to Y$, $B \to Y$: $\lambda_{ay} = r_{ya \cdot b}$, $\lambda_{by} = r_{yb \cdot a}$.

Example 4.5

$X \to Y$, $X \to W$, $Y \to W$, $Y \to Z$, $W \to Z$, $X \leftrightarrow Z$, $Y \leftrightarrow Z$: $\lambda_{xy} = r_{yx}$ ($\emptyset$ separates X, Y once $X \to Y$ is removed, since all other paths meet a collider), and $\lambda_{wz} = r_{zw \cdot yx}$ ($\{Y, X\}$ blocks $W \leftarrow Y \to Z$, $W \leftarrow Y \leftrightarrow Z$, $W \leftarrow X \leftrightarrow Z$).

4.2.6 Back-door criterion: total effects

Theorem 4.3 (Back-door, linear; Pearl 2000)

The total effect of X on Y is identifiable if there is $\mathbf{Z}$ with no descendant of X that d-separates X from Y in $\mathcal{G}_{\underline{X}}$; it equals $r_{yx \cdot \mathbf{Z}}$.

Proof [Supplement]. In a linear Gaussian model $\mathbb{E}[Y \mid \mathrm{do}(x), \mathbf{z}]$ is linear in x with slope equal to the total effect τ (sum of products of coefficients along directed paths). By the back-door adjustment theorem (Chapter 2), $\mathbb{E}[Y \mid \mathrm{do}(x)] = \sum_{\mathbf{z}} \mathbb{E}[Y \mid x, \mathbf{z}] P(\mathbf{z})$ (an integral in the continuous case). In the Gaussian case $\mathbb{E}[Y \mid x, \mathbf{z}] = \alpha + r_{yx \cdot \mathbf{Z}} x + \boldsymbol{\beta}^{\mathsf{T}} \mathbf{z}$, so $\mathbb{E}[Y \mid \mathrm{do}(x)] = \alpha + r_{yx \cdot \mathbf{Z}} x + \boldsymbol{\beta}^{\mathsf{T}} \mathbb{E}\mathbf{Z}$, whose slope in x is $r_{yx \cdot \mathbf{Z}}$. Hence $\tau = r_{yx \cdot \mathbf{Z}}$. ■

Example 4.6

$W \to X$, $W \leftrightarrow X$, $W \to Y$, $X \to Z \to Y$, $X \to Y$, $Z \leftrightarrow Y$. The total effect is $\lambda_{xz}\lambda_{zy} + \lambda_{xy}$; by the back-door criterion with $\{W\}$ it equals $r_{yx \cdot w}$. (The coefficient λ_{xy} alone is not obtainable by regression on Z because $Z \leftrightarrow Y$.)

4.3 Algorithmic identification methods

Motivation. View identification as solving the polynomial system linking Σ to (λ, ϵ).

4.3.1 The equations of linear identification

For $Z \xrightarrow{\lambda_{zx}} X \xrightarrow{\lambda_{xy}} Y$, $Z \leftrightarrow Y$:

$$\Sigma = \underbrace{\begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{bmatrix}}_{\text{computable}} = \underbrace{\begin{bmatrix} 1 & \lambda_{xy} + \lambda_{zx}\epsilon_{zy} & \lambda_{zx} \\ \lambda_{xy} + \lambda_{zx}\epsilon_{zy} & 1 & \lambda_{zx}\lambda_{xy} + \epsilon_{zy} \\ \lambda_{zx} & \lambda_{zx}\lambda_{xy} + \epsilon_{zy} & 1 \end{bmatrix}}_{\text{expressions with unobservables}}.$$

Can λ_{xy} be solved in terms of σ? Here $\sigma_{xz} = \lambda_{zx}$, and

$$\sigma_{xy} = \lambda_{xy} + \sigma_{xz}\epsilon_{zy}, \qquad \sigma_{zy} = \sigma_{xz}\lambda_{xy} + \epsilon_{zy}$$

are two full-rank linear equations in two unknowns (Gaussian elimination / row reduction):

$$\lambda_{xy} = \frac{\sigma_{xy} - \sigma_{xz}\sigma_{zy}}{1 - \sigma_{xz}^2}, \qquad \epsilon_{zy} = \frac{\sigma_{zy} - \sigma_{xz}\sigma_{xy}}{1 - \sigma_{xz}^2}.$$

Non-identifiability. For $X \to Y$ with $X \leftrightarrow Y$: $\sigma_{xy} = \lambda_{xy} + \epsilon_{xy}$ is one equation in two unknowns; infinitely many $(\lambda_{xy}, \epsilon_{xy})$ give the same covariance matrix.

Finite identifiability. $X \to W$, $X \to Z$, $X \to Y$ each also confounded with X ($X \leftrightarrow W$, $X \leftrightarrow Z$, $X \leftrightarrow Y$). Wright's rules give six equations in six unknowns:

$$\begin{aligned} \sigma_{xw} &= \lambda_{xw} + \epsilon_{xw}, & \sigma_{wz} &= \lambda_{xw}\lambda_{xz} + \lambda_{xz}\epsilon_{xw} + \lambda_{xw}\epsilon_{xz}, \\ \sigma_{xz} &= \lambda_{xz} + \epsilon_{xz}, & \sigma_{wy} &= \lambda_{xw}\lambda_{xy} + \lambda_{xw}\epsilon_{xy} + \lambda_{xy}\epsilon_{xw}, \\ \sigma_{xy} &= \lambda_{xy} + \epsilon_{xy}, & \sigma_{zy} &= \lambda_{xz}\lambda_{xy} + \lambda_{xz}\epsilon_{xy} + \lambda_{xy}\epsilon_{xz}. \end{aligned}$$

Eliminating: substitute $\epsilon = \sigma - \lambda$ to get $\sigma_{wz} = \lambda_{xz}\sigma_{xw} + \lambda_{xw}\sigma_{xz} - \lambda_{xw}\lambda_{xz}$ (and similarly for σ_{wy}, σ_{zy}); solve the last two for $\lambda_{xw} = (\sigma_{wy} - \lambda_{xy}\sigma_{xw})/(\sigma_{xy} - \lambda_{xy})$ and $\lambda_{xz} = (\sigma_{zy} - \lambda_{xy}\sigma_{xz})/(\sigma_{xy} - \lambda_{xy})$; plug into the first and clear denominators. The result is a quadratic in λ_{xy}:

$$0 = (\sigma_{xw}\sigma_{xz} - \sigma_{wz})\lambda_{xy}^2 + 2(\sigma_{xy}\sigma_{wz} - \sigma_{xw}\sigma_{xz}\sigma_{xy})\lambda_{xy} + (\sigma_{yw}\sigma_{xz}\sigma_{xy} + \sigma_{yz}\sigma_{xy}\sigma_{xw} - \sigma_{xy}^2\sigma_{wz} - \sigma_{yz}\sigma_{yw}),$$

so

$$\lambda_{xy} = \frac{-(\sigma_{xy}\sigma_{wz} - \sigma_{xw}\sigma_{xz}\sigma_{xy}) \pm \sqrt{(\sigma_{xy}\sigma_{wz} - \sigma_{xw}\sigma_{xz}\sigma_{xy})^2 - (\sigma_{xw}\sigma_{xz} - \sigma_{wz})\,c}}{\sigma_{xw}\sigma_{xz} - \sigma_{wz}},$$

with c the constant term. At most two values of λ_{xy} are consistent with the data: *finite identifiability*.

The three cases. *Identifiable*: a single value of λ_{xy} is consistent with the data. *Finitely identifiable*: finitely many values. *Not identifiable*: infinitely many.

Example 4.7

$W \to Z$, $W \to X$, $Z \to X$, $Z \leftrightarrow X$, $X \to Y$, $W \leftrightarrow Y$:

$$\begin{aligned} \sigma_{wz} &= \lambda_{wz}, & \sigma_{wy} &= (\lambda_{wx} + \lambda_{wz}\lambda_{zx})\lambda_{xy} + \epsilon_{wy}, \\ \sigma_{wx} &= \lambda_{wx} + \lambda_{wz}\lambda_{zx}, & \sigma_{zy} &= (\lambda_{zx} + \epsilon_{zx} + \lambda_{wz}\lambda_{wx})\lambda_{xy} + \lambda_{wz}\epsilon_{wy}, \\ \sigma_{zx} &= \lambda_{zx} + \epsilon_{zx} + \lambda_{wz}\lambda_{wx}, & \sigma_{xy} &= (\lambda_{wz}\lambda_{zx} + \lambda_{wx})\epsilon_{wy} + \lambda_{xy}. \end{aligned}$$

(Here $\lambda_{xy} = (\sigma_{zy} - \sigma_{wz}\sigma_{wy})/(\sigma_{zx} - \sigma_{wz}\sigma_{wx})$: Z is an instrument conditional on W, see below.) The goal of an ID algorithm is to find solvable subsystems efficiently; generic computer algebra is doubly exponential in the number of parameters.

4.4 From instrumental variables to instrumental sets

Motivation. Use variables that affect Y only through X to build solvable linear subsystems.

4.4.1 Instrumental variables

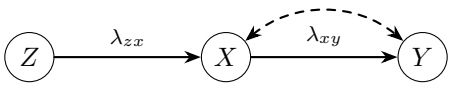


Figure. *The instrumental-variable graph. Confounding ($X \leftrightarrow Y$) makes $\sigma_{xy} \neq \lambda_{xy}$, but Z has no trek to Y except through X, so $\sigma_{zy} = \lambda_{zx}\lambda_{xy}$ and $\sigma_{zx} = \lambda_{zx}$ give $\lambda_{xy} = \sigma_{zy}/\sigma_{zx}$.*

Suppose an independent doctor's recommendation Z, which influences how much drug patients take, is added: $Z \xrightarrow{\lambda_{zx}} X \xrightarrow{\lambda_{xy}} Y$, $X \leftrightarrow Y$. (Non-parametrically, $P(y \mid \mathrm{do}(x))$ is *not* identifiable here; linearity is what helps.)

Definition (Instrumental variable; Pearl 2000, p. 248)

Z is an IV for λ_{xy} if Z is d-separated from Y in $\mathcal{G}_{\lambda_{xy}}$ and Z is not d-separated from X in $\mathcal{G}_{\lambda_{xy}}$.

By Wright's rules $\sigma_{zx} = \lambda_{zx}$ and $\sigma_{zy} = \lambda_{zx}\lambda_{xy}$, so

$$\lambda_{xy} = \frac{\sigma_{zy}}{\sigma_{zx}} = \frac{r_{zy}}{r_{zx}}.$$

Two-stage least squares. Regress $X = \alpha Z + e$, giving $\alpha = \sigma_{zx}$; set $\hat{X} = \sigma_{zx} Z$ and regress $Y = \beta \hat{X} + e$, giving $\beta = \sigma_{zy}/\sigma_{zx}$ (since $\mathrm{Cov}(Y, \hat{X}) = \sigma_{zx}\sigma_{zy}$ and $\mathrm{Var}(\hat{X}) = \sigma_{zx}^2$).

4.4.2 Conditional instrumental variables

Definition (Conditional IV; Brito and Pearl 2002)

Z is a conditional IV given $\mathbf{W}$ for λ_{xy} if (i) $\mathbf{W}$ contains only non-descendants of Y;
(ii) $\mathbf{W}$ d-separates Z from Y in $\mathcal{G}_{\lambda_{xy}}$;
(iii) $\mathbf{W}$ does not d-separate Z from X in $\mathcal{G}_{\lambda_{xy}}$.

Then $\lambda_{xy} = r_{yz \cdot \mathbf{W}} / r_{xz \cdot \mathbf{W}}$. Example: $W \to Z \to X \to Y$, $X \leftrightarrow Y$, $W \leftrightarrow Y$. Z is not an IV ($Z \leftarrow W \leftrightarrow Y$ is open) but is a conditional IV given W.

4.4.3 Instrumental sets

$Z_1 \to X_1$, $Z_1 \to X_2$, $Z_2 \to X_1$, $Z_2 \to X_2$, $X_1 \to Y$, $X_2 \to Y$, $X_1 \leftrightarrow Y$, $X_2 \leftrightarrow Y$. No single IV exists, since every path from Z_1 or Z_2 to Y passes through both X_1 and X_2. How to solve for $\lambda_{x_1 y}$? We have $\sigma_{z_i x_j} = \lambda_{z_i x_j}$ for $i, j \in \{1, 2\}$ and

$$\sigma_{z_1 y} = \sigma_{z_1 x_1}\lambda_{x_1 y} + \sigma_{z_1 x_2}\lambda_{x_2 y}, \qquad \sigma_{z_2 y} = \sigma_{z_2 x_1}\lambda_{x_1 y} + \sigma_{z_2 x_2}\lambda_{x_2 y},$$

a linear system built from paths from $\{Z_1, Z_2\}$ to $\{X_1, X_2\}$, solvable by Cramer's rule if its matrix is non-singular.

When does the system fail? The equations must not be collinear. If $Z_1 \to Z_2 \to \{X_1, X_2\}$, then paths from Z_1 to Y are paths from Z_2 to Y multiplied by $\lambda_{z_1 z_2}$:

$$\lambda_{z_1 z_2}\sigma_{z_2 y} = \lambda_{z_1 z_2}(\sigma_{z_2 x_1}\lambda_{x_1 y} + \sigma_{z_2 x_2}\lambda_{x_2 y}), \qquad \sigma_{z_2 y} = \sigma_{z_2 x_1}\lambda_{x_1 y} + \sigma_{z_2 x_2}\lambda_{x_2 y}.$$

The two equations are identical up to a constant factor: the system is degenerate.

Theorem 4.4 (Non-intersecting paths)

The system is non-degenerate (generically) iff there is a full matching between $\{Z_1, Z_2\}$ and $\{X_1, X_2\}$ by vertex-disjoint (non-intersecting) paths.

Proof [Supplement]. For the 2×2 system the instrumental equations are $\sigma_{z_i y} = \sum_j \sigma_{z_i x_j}\lambda_{x_j y}$, $i = 1, 2$, which have a unique solution iff $\det[\sigma_{z_i x_j}] \neq 0$. By Wright's rules each $\sigma_{z_i x_j}$ is a sum over treks, and the Lindström–Gessel–Viennot lemma expresses the determinant as a signed sum over pairs of vertex-disjoint path systems from $\{Z_1, Z_2\}$ to $\{X_1, X_2\}$: if no vertex-disjoint system exists, all terms cancel and the determinant is identically zero; if one exists, the determinant is a nonzero polynomial in the coefficients and hence nonzero for generic parameter values. (The general $k \times k$ statement is Brito and Pearl 2002.) ■

In the first graph, $Z_1 \to X_1$ and $Z_2 \to X_2$ are disjoint; in the degenerate graph, both paths must pass through Z_2. Such paths can be found by max-flow: treat each Z_i as a source and each X_j as a sink with unit vertex capacities; a flow of value $|\mathbf{Z}|$ certifies a full matching by non-intersecting paths.

4.4.4 Conclusions

1. Linear SCMs are the most common causal models in real-world applications.
2. Be careful with regression analysis and with what it actually computes.
3. A structural equation is not a regression equation, but regression sometimes recovers structural coefficients.
4. The causal diagram can be leveraged to solve for causal effects in many practical and general settings.

4.5 Appendix: Complexity of identification in linear SCMs

Landscape. Single-door, IV and instrumental sets are sufficient but incomplete. Other graphical criteria (half-trek criterion, auxiliary cutsets/ACID; Chen et al. 2016, Kumor et al. 2019) cover more instances, but not all identifiable ones. Is the general problem harder than these criteria?

Limits. The graphical criteria run in polynomial time but miss some identifiable instances. Brute-force computer algebra (Gröbner bases) solves all of them but takes doubly exponential time; a 4-node graph can take over 75 days (García-Puente, Spielvogel and Sullivant 2010; Bardet 2002).

Recent result (Dörfler, van der Zander, Bläser and Liśkiewicz, NeurIPS 2024). The general identification problem is provably hard, harder than NP, because it requires reasoning over real-valued unknowns rather than binary ones. Nevertheless it can be solved in polynomial space (PSPACE), hence in at most $2^{\mathrm{poly}(n)}$ time, a large improvement over the $2^{2^{\mathrm{poly}(n)}}$ of Gröbner bases. Take-away: when graphical criteria work, use them; otherwise the problem is inherently difficult, but not impossible.

4.6 Appendix: Multiple regression when Y is the cause

$X_1 \xleftarrow{b_1} Y \xrightarrow{b_2} X_2$. Then $\sigma_{yx_1} = b_1$, $\sigma_{yx_2} = b_2$, $\sigma_{x_1x_2} = b_1b_2$. The normal equations of $Y = \beta_1 X_1 + \beta_2 X_2 + e$ are

$$b_1 = \beta_1 + \beta_2 b_1 b_2, \qquad b_2 = \beta_2 + \beta_1 b_1 b_2.$$

Substituting $\beta_2 = b_2 - \beta_1 b_1 b_2$ into the first: $b_1 = \beta_1 + b_1 b_2^2 - \beta_1 b_1^2 b_2^2$, so $\beta_1(1 - b_1^2 b_2^2) = b_1(1 - b_2^2)$ and

$$\beta_1 = \frac{b_1(1 - b_2^2)}{1 - b_1^2 b_2^2}, \qquad \beta_2 = \frac{b_2(1 - b_1^2)}{1 - b_1^2 b_2^2}.$$

4.7 Appendix: The Gröbner basis approach

Identification as polynomial elimination. By Wright's rules, observed covariances are polynomials in the unknowns λ, ϵ. Example: $X_1 \xrightarrow{\lambda_{12}} X_2 \xrightarrow{\lambda_{23}} X_3$ with $X_2 \leftrightarrow X_3$ (ϵ_{23}):

$$\sigma_{12} = \lambda_{12}, \qquad \sigma_{13} = \lambda_{12}\lambda_{23}, \qquad \sigma_{23} = \lambda_{23} + \epsilon_{23}.$$

From the first two, $\lambda_{23} = \sigma_{13}/\sigma_{12}$ (X_1 is an instrument); from σ_{23} alone we could not separate λ_{23} from ϵ_{23}. Gröbner bases automate this "solve and eliminate" process for any graph.

How it works (intuition).

1. *Set up:* write $\sigma_{ij} -$ (polynomial in $\lambda, \epsilon) = 0$ for every pair (i, j).
2. *Eliminate:* use elimination term orders to remove unwanted unknowns, keeping only the target (say λ_{xy}) and the σ_{ij}.
3. *Check:* a degree-1 polynomial in λ_{xy} means generically identifiable; degree d means algebraically d-identifiable (d possible values; may or may not yield unique identification); if λ_{xy} does not appear, it is not generically identifiable.

Computational results (García-Puente et al. 2010; all linear SCMs on 3 and 4 nodes):

	3 nodes ($2^6 = 64$)	4 nodes ($2^{12} = 4096$)
Generically identifiable	22	1246
Not generically identifiable	42	2844
Algebraically k-identified	0	6
Solved by single-door / IV	22/22	1093/1246
Only by Gröbner basis	0	153

On 3 nodes the graphical criteria suffice; on 4 nodes 153 cases need the algebraic method. Runtimes range from seconds to more than 75 days for a single 4-node graph.